

A Maven Repository Download URL Construction

```
https://<repository>/<package>/<dependency>/<version>
/<dependency>--<version>.<jar|aar>
# Example:
https://repo1.maven.org/maven2/org/slf4j/slf4j-api/1.1.0
/slf4j-api-1.1.0.jar
```

Listing 17: General structure of Maven repository download URLs.

```
https://repo1.maven.org/maven2
https://repo.clojars.org
https://repo.akka.io/maven
https://maven.google.com
https://maven.artifacts.atlassian.com
https://maven.wso2.org/nexus/content/repositories/releases
https://nexus.bedatadriven.com/content/groups/public
https://repository.mulesoft.org/nexus/content/repositories/public
https://repo.jenkins-ci.org/releases
https://nexus.senbox.net/nexus/content/repositories/releases
https://open.artefacts.tax.service.gov.uk/maven2
```

Table 8: Maven repository base URLs.

B Sink Method Filtering for Gadget Chain Verification

Sink Method	Count
java.lang.reflect.Method.invoke	608
java.lang.ClassLoader.defineClass	32
org.springframework.jndi.JndiTemplate.lookup	3
java.sql.PreparedStatement.execute	1
java.io.FileOutputStream.write	168
java.lang.ClassLoader.loadClass	77
java.net.URL.openConnection	142
java.sql.Statement.execute	1
javax.naming.InitialContext.lookup	1
java.lang.reflect.Constructor.newInstance	423
java.io.FileOutputStream.<init>	91
java.sql.PreparedStatement.executeQuery	3
java.io.File.delete	91
java.beans.Introspector.getBeanInfo	3
java.net.URL.openStream	119
java.sql.DriverManager.getConnection	1
java.sql.Connection.prepareStatement	4
java.nio.file.Files.newOutputStream	28
javax.naming.Context.lookup	900
java.lang.ProcessBuilder.<init>	11
java.lang.Runtime.exec	18
java.rmi.registry.Registry.lookup	1
java.nio.file.Files.newBufferedWriter	7
<i>Filtered</i>	
java.io.FileInputStream.<init>	202
java.lang.Class.forName	239
java.lang.Class.getMethod	260
java.lang.Class.getDeclaredMethod	48
java.lang.ClassLoader.getResourceAsStream	12
java.lang.Class.getResourceAsStream	5
java.nio.file.Files.readAllLines	7
java.io.FileReader.<init>	1
org.xml.sax.XMLReader.parse	4
java.io.RandomAccessFile.read	4
java.io.RandomAccessFile.readFully	2
java.nio.file.Files.newInputStream	81
java.nio.file.Files.newBufferedReader	5
java.nio.file.Files.readAllBytes	9
java.util.zip.ZipInputStream.<init>	1
com.esotericsoftware.kryo.Kryo.readClassAndObject	2

Table 9: Sink methods in dormant gadget chains. The sink methods in the lower section are filtered out for the manual verification. Note that we also disregard the dynamically identified sinks by *Crystallizer*. The definition of considering any method which operates on arbitrary objects produces a high amount of false positive sink methods [65].

C Verified Dormant Gadget Chains

C.1 Transitive Serializability

```
java.util.Hashtable.readObject()
↳ java.util.Hashtable.readHashtable()
↳ java.util.Hashtable.reconstitutionPut()
↳ com.github.jknack.handlebars.io.ReloadableTemplateSource.equals()
↳ com.github.jknack.handlebars.io.URLTemplateSource.lastModified()
↳ com.github.jknack.handlebars.io.URLTemplateSource(URL)
↳ java.lang.reflect.Method.invoke()
↳ java.net.URL.openConnection()
```

Table 10: *com.github.jknack.handlebars-4.4.0*

C.2 Final Attributes

```
org.apache.flink.api.common.state.StateDescriptor.readObject()
↳ org.apache.flink.api.java.typeutils.runtime.kryo.KryoSerializer.deserialize()
↳ org.apache.flink.api.java.typeutils.runtime.kryo.KryoSerializer.checkKryoInitialized()
↳ org.apache.flink.api.java.typeutils.runtime.kryo.KryoSerializer.getKryoInstance()
↳ java.lang.reflect.Method.invoke()
```

Table 11: *org.apache.flink.flink-core-1.19.2*

```
org.jfree.chart.plot.CombinedRangeCategoryPlot.readObject()
↳ org.jfree.chart.axis.PeriodAxis.configure()
↳ org.jfree.chart.axis.PeriodAxis.autoAdjustRange()
↳ org.jfree.chart.axis.PeriodAxis.createInstance()
↳ java.lang.reflect.Constructor.newInstance()
```

Table 12: *org.jfree.jfreechart-1.5.5*

```
org.rogach.scallop.SerializationProxy.readObject()
↳ java.lang.reflect.Constructor.newInstance()
```

Table 13: *org.rogach.scallop-5.2.0*

C.3 Interface Method Reachability

```
java.lang.Runnable.run()
↳ org.springframework.beans.factory.support.DisposableBeanAdapter.run()
↳ org.springframework.beans.factory.support.DisposableBeanAdapter.destroy()
↳ org.springframework.beans.factory.support.DisposableBeanAdapter.invokeCustomDestroyMethod()
↳ java.lang.reflect.Method.invoke()
```

Table 14: *org.springframework.spring-beans-6.1.17*

```
java.beans.PropertyChangeListener.propertyChange()
↳ org.htmlparser.beans.HTMLTextBean.propertyChange()
↳ javax.swing.text.JTextComponent.setText()
↳ javax.swing.text.AbstractDocument.replace()
↳ javax.swing.text.DefaultFormatter$DefaultDocumentFilter.replace()
↳ javax.swing.text.NumberFormatter.replace()
↳ javax.swing.text.NumberFormatter.toggleSignIfNecessary()
↳ javax.swing.text.NumberFormatter.toggleSign()
↳ java.lang.reflect.Constructor.newInstance()
```

Table 15: *org.htmlparser.htmlparser-2.1*

```
javax.sql.DataSource.getConnection()
↳ org.apache.commons.dbcp2.datasources.InstanceKeyDataSource.getConnection()
↳ org.apache.commons.dbcp2.datasources.InstanceKeyDataSource.testCPDS()
↳ javax.naming.InitialContext.lookup()
↳ javax.naming.Context.lookup()
```

Table 16: *org.apache.commons.commons-dbcp2-2.13.0*

```
java.lang.AutoCloseable.close()
↳ org.mapdb.BTreeMap.close()
↳ org.mapdb.StoreTrivialTx.close()
↳ java.io.File.delete()
```

Table 17: *org.mapdb.mapdb-3.1.0*

```
java.lang.reflect.InvocationHandler.invoke()
↳ org.apache.openejb.threads.impl.ContextServiceImpl$CUHandler.invoke()
↳ org.apache.openejb.threads.impl.ContextServiceImpl$CUHandler.lambda$invoke$0()
↳ java.lang.reflect.Method.invoke()
```

Table 18: *org.apache.tomee.openejb-core-10.0.0*

```
java.lang.Runnable.run()
↳ com.googlecode.aviator.runtime.function.AbstractVariadicFunction.run()
↳ com.googlecode.aviator.runtime.function.AbstractVariadicFunction.call()
↳ com.googlecode.aviator.runtime.function.ClassMethodFunction.variadicCall()
↳ com.googlecode.aviator.util.Reflector.invokeStaticMethod()
↳ com.googlecode.aviator.util.Reflector.invokeMatchingMethod()
↳ java.lang.reflect.Method.invoke()
```

Table 19: *com.googlecode.aviator.aviator-5.4.2*

```
javax.sql.RowSet.rollback()
↳ oracle.jdbc.rowset.OracleCachedRowSet.rollback()
↳ oracle.jdbc.rowset.OracleCachedRowSet.getConnectionInternal()
↳ javax.naming.InitialContext.lookup()
↳ javax.naming.Context.lookup()
```

Table 20: *com.oracle.database.jdbc.ojdbcx-21.17.0.0* - this gadget chain is the same for ojdbc 11, 8 and 6

```
java.lang.reflect.InvocationHandler.invoke()
↳ org.hibernate.validator.internal.util.annotation.AnnotationProxy.invoke()
↳ java.lang.reflect.Method.invoke()
```

Table 21: *org.hibernate.validator.hibernate-validator-8.0.2*

```
java.util.concurrent.Callable.call()
↳ org.redisson.mapreduce.CoordinatorTask.call()
↳ java.lang.reflect.Constructor.newInstance()
```

Table 22: *org.redisson.redisson-3.44.0*

```
java.lang.Runnable.run()
↳ weka.gui.experiment.RunPanel$ExperimentRunner.run()
↳ weka.experiment.Experiment.initialize()
↳ weka.experiment.RandomSplitResultProducer.preProcess()
↳ weka.experiment.CSVResultListener.preProcess()
↳ java.io.FileOutputStream.<init>()
```

Table 23: *nz.ac.waikato.cms.weka.weka-dev-3.9.6*

```
java.lang.reflect.InvocationHandler.invoke()
↳ org.apache.bval.jsr.util.AnnotationProxy.invoke()
↳ java.lang.reflect.Method.invoke()
```

Table 24: *org.apache.bval.bval-jsr-3.0.1*

```
java.awt.event.ActionListener.actionPerformed()
↳ org.jdesktop.swingx.action.ServerAction.actionPerformed()
↳ java.net.URL.openConnection()
```

Table 25: *org.swinglabs.swingx-1.6.1*

```
java.lang.reflect.InvocationHandler.invoke()
↳ org.apache.ibatis.logging.jdbc.ResultSetLogger.invoke()
↳ java.lang.reflect.Method.invoke()
```

Table 26: *org.mybatis.mybatis-3.5.19*

```
javax.sql.XADataSource.getXAConnection()
↳ com.alibaba.druid.pool.xa.getXAConnection()
↳ com.alibaba.druid.pool.DruidDataSource.getConnection()
↳ com.alibaba.druid.pool.DruidDataSource.getConnection()
↳ com.alibaba.druid.pool.DruidDataSource.getConnectionDirect()
↳ com.alibaba.druid.pool.DruidDataSource.getConnectionInternal()
↳ com.alibaba.druid.pool.DruidDataSource.takeLast()
↳ com.alibaba.druid.pool.DruidDataSource.pollLast()
↳ com.alibaba.druid.pool.DruidDataSource$CreateConnectionThread.run()
↳ com.alibaba.druid.pool.DruidAbstractDataSource.createPhysicalConnection()
↳ com.alibaba.druid.pool.DruidAbstractDataSource.createPhysicalConnection()
↳ java.sql.Driver.connect()
```

Table 27: *com.alibaba.druid-1.2.24*

```
java.lang.reflect.InvocationHandler.invoke()
↳ org.springframework.core.SerializableTypeWrapper$TypeProxyInvocationHandler.invoke()
↳ org.springframework.util.ReflectionUtils.invokeMethod()
↳ java.lang.reflect.Method.invoke()
```

Table 28: *org.springframework.spring-core-6.1.17*

```
javax.naming.Context.lookup()
↳ org.apache.activemq.jndi.ReadOnlyContext.lookup()
↳ javax.naming.Context.lookup()
```

Table 29: *org.apache.activemq.activemq-client-6.1.5*. The interesting part is that `ReadOnlyContext` is serializable

```
javax.naming.Context.lookup()
↳ org.apache.camel.util.jndi.JndiContext.lookup()
↳ javax.naming.Context.lookup()
```

Table 30: *org.apache.camel.camel-core-2.25.4*

```
javax.sql.DataSource.getConnection()
↳ net.sourceforge.jtds.jdbcx.JtdsDataSource.getConnection()
↳ net.sourceforge.jtds.jdbcx.JtdsDataSource.getConnection(String, String)
↳ java.io.FileOutputStream.<init>()
```

Table 31: *net.sourceforge.jtds.jtds-1.2.8*

```
java.lang.reflect.InvocationHandler.invoke()
↳ io.lettuce.core.dynamic.support.TypeWrapper$TypeProxyInvocationHandler.invoke()
↳ java.lang.reflect.Method.invoke()
```

Table 32: *io.lettuce.lettuce-core-6.5.4*

```

java.sql.DataSource.getConnection()
↪ org.apache.derby.client.BasicClientDataSource.getConnection()
↪ org.apache.derby.client.BasicClientDataSource.computeDnLogWriterForNewConnection()
↪ org.apache.derby.client.BasicClientDataSource.computeDnLogWriter()
↪ org.apache.derby.client.BasicClientDataSource.computePrintWriter()
↪ org.apache.derby.client.BasicClientDataSource.getPrintWriter()
↪ java.io.FileOutputStream.<init>()

```

Table 33: *org.apache.derby.derbyclient-10.17.1.0*

```

javax.xml.transform.Transformer newTransformer()
↪ com.sun.org.apache.xalan.internal.xsltc.trax.TemplatesImpl.newTransformer()
↪ com.sun.org.apache.xalan.internal.xsltc.trax.TemplatesImpl.getTransletInstance()
↪ java.lang.reflect.Constructor.newInstance()

```

Table 34: *xalan.xalan-2.7.3*

C.4 Mixed

```

java.lang.Runnable.run()
↪ org.apache.hadoop.security.UserGroupInformation$AutoRenewalForUserCredsRunnable.run()
↪ org.apache.hadoop.security.UserGroupInformation$TicketCacheRenewalRunnable.relogin()
↪ org.apache.hadoop.util.Shell.execCommand(java.lang.String[])
↪ org.apache.hadoop.util.Shell.execCommand(java.util.Map, java.lang.String[], long)
↪ org.apache.hadoop.util.Shell$ShellCommandExecutor.execute()
↪ org.apache.hadoop.util.Shell.run()
↪ org.apache.hadoop.util.Shell.runCommand()
↪ java.lang.ProcessBuilder.<init>()

```

Table 35: *org.apache.hadoop.hadoop-common-3.4.1*

```

java.util.Hashtable.readObject()
↪ java.util.Hashtable.readHashtable()
↪ java.util.Hashtable.reconstitutionPut()
↪ java.net.URL.equals()
↪ org.eclipse.osgi.internal.url.MultiplexingURLStreamHandler.equals()
↪ org.eclipse.osgi.internal.url.MultiplexingURLStreamHandler.findAuthorizedURLStreamHandler()
↪ org.eclipse.osgi.internal.url.URLStreamHandlerFactoryImpl.findAuthorizedURLStreamHandler()
↪ java.lang.reflect.Method.invoke()

```

Table 36: *org.eclipse.platform.org.eclipse.osgi-3.22.0*

```

java.lang.AutoCloseable.close()
↪ java.util.Hashtable.readHashtable()
↪ org.apache.poi.openxml4j.opc.OPCPackage.close()
↪ org.apache.poi.openxml4j.opc.ZipPackage.closeImpl()
↪ org.apache.poi.openxml4j.opc.OPCPackage.save()
↪ java.nio.file.Files.newOutputStream()

```

Table 37: *org.apache.poi.poi-ooxml-5.4.0*

```

java.awt.event.ActionListener.actionPerformed()
↪ org.h2.tools.GUIConsole.actionPerformed()
↪ org.h2.tools.GUIConsole.shutdown()
↪ org.h2.util.Utils.callMethod()
↪ org.h2.util.Utils.callMethod()
↪ java.lang.reflect.Method.invoke()

```

Table 38: *com.h2database.h2-2.3.232*

```

java.lang.AutoCloseable.close()
↪ org.apache.hadoop.fs.store.DataBlocks$DataBlock.close()
↪ org.apache.hadoop.fs.store.DataBlocks$DiskBlock.innerClose()
↪ org.apache.hadoop.fs.store.DataBlocks$DiskBlock.closeBlock()
↪ java.io.File.delete()

```

Table 39: *org.apache.hadoop.hadoop-common-3.4.1*

```

java.lang.AutoCloseable.close()
↪ org.apache.cassandra.io.util.RewindableDataInputStreamPlus.close()
↪ org.apache.cassandra.io.util.RewindableDataInputStreamPlus.close(boolean)
↪ java.io.File.delete()

```

Table 40: *org.apache.cassandra.cassandra-all-3.0.32*

```

java.lang.Runnable.run()
↪ org.apache.logging.log4j.core.appender.rolling.action.AbstractAction.run()
↪ org.apache.logging.log4j.core.appender.rolling.action.ZipCompressAction.execute()
↪ org.apache.logging.log4j.core.appender.rolling.action.ZipCompressAction.execute()
↪ java.io.File.delete()

```

Table 41: *org.apache.log4j.log4j-core-2.24.3*

```

java.util.Iterator.hasNext()
↪ org.apache.openjpa.jdbc.meta.strats.LRSPProxyMap$ResultIterator.hasNext()
↪ org.apache.openjpa.jdbc.sql.MergedResult.next()
↪ org.apache.openjpa.jdbc.sql.LogicalUnion$ResultComparator.getOrderingValue(Result, int)
↪ org.apache.openjpa.jdbc.sql.LogicalUnion$ResultComparator.getOrderingValue(Result, Object)
↪ org.apache.openjpa.jdbc.sql.PostgresDictionary.getObject()
↪ java.lang.reflect.Method.invoke()

```

Table 42: *org.apache.openjpa.openjpa-4.0.1*

```

java.lang.Runnable.run()
↪ org.jgroups.protocols.MERGE3$InfoSender.run()
↪ org.jgroups.protocols.PDC.down()
↪ org.jgroups.protocols.PDC.writeNodeToDisk()
↪ org.jgroups.protocols.PDC.writeToTempFile()
↪ java.io.FileOutputStream.<init>()

```

Table 43: *org.jgroups.jgroups-5.3.15*

```

java.beans.PropertyChangeListener.propertyChange()
↪ org.apache.catalina.core.NamingContextListener.propertyChange()
↪ org.apache.catalina.core.NamingContextListener.processGlobalResourcesChange()
↪ org.apache.catalina.core.NamingContextListener.addEnvironment()
↪ org.apache.catalina.core.NamingContextListener.constructEnvEntry()
↪ java.lang.reflect.Constructor.newInstance()

```

Table 44: *org.apache.tomcat.embed.tomcat-embed-core-10.1.36* and *org.apache.tomcat.tomcat-catalina-10.1.36*

```

java.util.Iterator.hasNext()
↪ org.python.core.WrappedIterIterator.hasNext()
↪ org.python.modules.itertools.inap$1.__iternext__()
↪ org.python.core.PyBuiltinMethodNarrow.__call__()
↪ org.python.core.PyJavaType$14.__call__()
↪ java.lang.reflect.Method.invoke()

```

Table 45: *org.python.jython-standalone-2.7.4*

```

java.lang.Runnable.run()
↪ net.bytebuddy.ClassFileVersion$VersionLocator$Resolver.run()
↪ java.lang.reflect.Method.invoke()

```

Table 46: *net.bytebuddy.byte-buddy-1.17.1*

```

java.util.Map.put()
↪ org.apache.commons.beanutils.BaseDynaBeanMapDecorator.put()
↪ org.apache.commons.beanutils.LazyDynaBean.get()
↪ org.apache.commons.beanutils.LazyDynaBean.createProperty()
↪ org.apache.commons.beanutils.LazyDynaBean.createOtherProperty()
↪ java.lang.reflect.Constructor.newInstance()

```

Table 47: *commons-beanutils-1.10.1*

```

java.io.Flushable.flush()
↪ org.springframework.integration.metadata.PropertiesPersistingMetadataStore.flush()
↪ org.springframework.integration.metadata.PropertiesPersistingMetadataStore.saveMetadata()
↪ java.io.FileOutputStream.<init>()

```

Table 48: *spring-integration-core-6.3.8*

```

java.util.Iterator.next()
↪ com.mysql.cj.xdevapi.AbstractDataResult.next()
↪ com.mysql.cj.protocol.ProtocolEntityFactory.createFromProtocolEntity()
↪ com.mysql.cj.exceptions.ExceptionFactory.createException()
↪ java.lang.reflect.Constructor.newInstance()

```

Table 49: *com.mysql.mysql-connector-j-9.2.0*

```

java.lang.AutoCloseable.close()
↪ org.apache.sshd.agent.unix.AgentServerProxy.close()
↪ org.apache.sshd.agent.unix.AgentServerProxy.removeSocketFile()
↪ org.apache.sshd.agent.unix.AgentServerProxy.deleteFile()
↪ java.io.File.delete()

```

Table 50: *org.apache.sshd.sshd-core-2.14.0*

```

java.lang.Runnable.run()
↪ org.apache.calcite.adapter.druid.DruidConnectionImpl$1.run()
↪ org.apache.calcite.adapter.druid.DruidConnectionImpl.request()
↪ org.apache.calcite.runtime.HttpUtils.post()
↪ org.apache.calcite.runtime.HttpUtils.executeMethod()
↪ org.apache.calcite.runtime.HttpUtils.getURLConnection()
↪ java.net.URL.openConnection()

```

Table 51: *org.apache.hive.hive-exec-4.0.1*

```

java.lang.reflect.InvocationHandler.invoke()
↪ org.codehaus.groovy.runtime.ConversionHandler.invoke()
↪ java.lang.reflect.Method.invoke()

```

Table 52: *org.apache.groovy.groovy-4.0.25*

```

java.lang.Iterable.iterator()
↪ com.google.javascript.jscomp.jarjar.org.apache.tools.ant.types.resources.Files.iterator()
↪ com.google.javascript.jscomp.jarjar.org.apache.tools.ant.types.ArchiveScanner.scan()
↪ com.google.javascript.jscomp.jarjar.org.apache.tools.ant.types.resources.URLResource.isExists()
↪ com.google.javascript.jscomp.jarjar.org.apache.tools.ant.types.resources.URLResource.isExists()
↪ com.google.javascript.jscomp.jarjar.org.apache.tools.ant.types.resources.URLResource.connect()
↪ java.net.URL.openConnection()

```

Table 53: *com.google.javascript.closure-compiler-20240317*

```

java.lang.Runnable.run()
↪ io.undertow.server.handlers.resource.URLResource$1ServerTask.run()
↪ java.net.URL.openStream()

```

Table 54: *io.undertow.undertow-core-2.2.37*

```
java.sql.Connection.isValid()
↳ com.clickhouse.jdbc.internal.ClickHouseConnectionImpl.isValid()
↳ com.clickhouse.client.AbstractClient.ping()
↳ com.clickhouse.client.http.ClickHouseHttpClient.checkHealth()
↳ com.clickhouse.client.http.HttpURLConnectionImpl.ping()
↳ com.clickhouse.client.http.HttpURLConnectionImpl.newConnection()
↳ java.net.URLConnection.openConnection()
```

Table 55: *com.clickhouse.clickhouse-jdbc-0.8.1*

```
java.lang.AutoCloseable.close()
↳ liquibase.database.core.DerbyDatabase.close()
↳ liquibase.database.core.DerbyDatabase.shutdownDerby()
↳ java.lang.reflect.Constructor.newInstance()
```

Table 56: *org.liquibase.liquibase-core-4.31.1*

```
java.util.Iterator.next()
↳ edu.stanford.nlp.objectbank.ReaderIteratorFactory$ReaderIterator.next()
↳ edu.stanford.nlp.objectbank.ReaderIteratorFactory$ReaderIterator.setNextObject()
↳ java.net.URL.openStream()
```

Table 57: *edu.stanford.nlp.stanford-corenlp-4.5.8*

```
java.util.Iterator.hasNext()
↳ org.eclipse.persistence.queries.ScrollableCursor.hasNext()
↳ org.eclipse.persistence.queries.ScrollableCursor.loadNext()
↳ org.eclipse.persistence.queries.ScrollableCursor.retrieveNextObject()
↳ org.eclipse.persistence.internal.queries.JoinedAttributeManager.processDataResults()
↳ org.eclipse.persistence.internal.descriptors.ObjectBuilder.extractPrimaryKeyFromRow()
↳ org.eclipse.persistence.internal.descriptors.ObjectBuilder.extractPrimaryKeyFromObject()
↳ org.eclipse.persistence.internal.descriptors.ObjectBuilder.extractPrimaryKeyFromObject()
↳ org.eclipse.persistence.mappings.structures.NestedTableMapping.writeFromObjectIntoRow()
↳ org.eclipse.persistence.internal.databaseaccess.DataSourceAccessor.incrementCallCount()
↳ org.eclipse.persistence.internal.databaseaccess.DataSourceAccessor.reconnect()
↳ org.eclipse.persistence.internal.databaseaccess.DataSourceAccessor.connectInternal()
↳ org.eclipse.persistence.sessions.DataSourceLogin.connectToDataSource()
↳ org.eclipse.persistence.sessions.JNDIConnector.connect()
↳ javax.naming.Context.lookup()
```

Table 58: *org.eclipse.persistence.eclipselink-4.0.5*

```
java.util.Iterator.hasNext()
↳ org.hibernate.query.internal.ScrollableResultsIterator.hasNext()
↳ org.hibernate.internal.FetchingScrollableResultsImpl.next()
↳ org.hibernate.internal.FetchingScrollableResultsImpl.prepareCurrentRow()
↳ org.hibernate.sql.results.internal.StandardReader.readRow()
↳ org.hibernate.sql.results.internal.RowTransformerTupleTransformerAdapter.transformRow()
↳ org.hibernate.jpa.spi.NativeQueryConstructorTransformer.transformTuple()
↳ java.lang.reflect.Constructor.newInstance()
```

Table 59: *org.hibernate.orm.hibernate-core-6.6.8*

D Apache OpenJPA Payload Construction

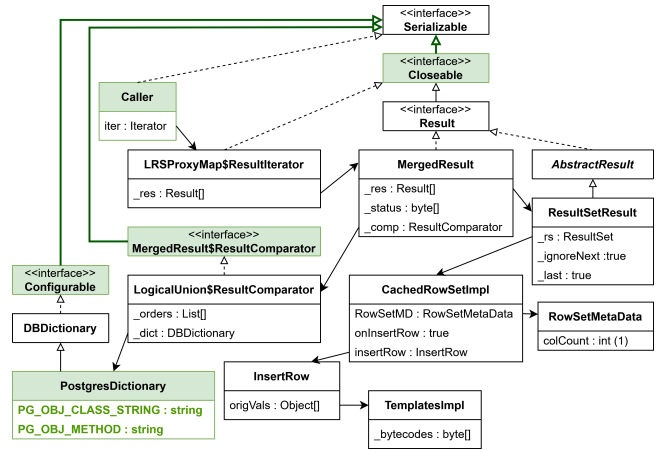


Figure 6: Payload visualization for dormant *OpenJPA* gadget chain. Modifications are highlighted in green. Making `LRSPProxyMap$ResultIterator`, `MergedResult`, and `ResultSetResult` serializable requires only the addition of *Serializable* to *OpenJPA*'s *Closeable* since all three gadgets inherit from it. It illustrates quite well how changing the serializability of a single interface (or class) can have a substantial impact on its descendants. Additionally, we provide transitive serializability via `Configurable` and `MergedResults$ResultComparator` to `PostgresDictionary` and `LogicalUnion$ResultComparator`, respectively.

E Deserialization Filter

```

jdk.serialFilter=
!com.github.jknack.handlebars.io.URLTemplateSource;
!org.jfree.chart.axis.PeriodAxis;
!org.apache.flink.api.java.typeutils.runtime.kryo.KryoSerializer;
!org.rogach.scallop.SerializationProxy;
!org.springframework.beans.factory.support.DisposableBeanAdapter;
!org.htmlparser.beans.HTMLTextBean;
!org.apache.commons.dbcp2.datasources.InstanceKeyDataSource;
!org.mapdb.StoreTrivialTx;
!org.apache.openejb.threads.impl.ContextServiceImpl$CUHandler;
!org.apache.openejb.threads.impl.ContextServiceImpl$CUHandler.lambda$invoke$0;
!com.googlecode.aviator.runtime.function.ClassMethodFunction;
!oracle.jdbc.rowset.OracleCachedRowSet;
!org.hibernate.validator.internal.util.annotation.AnnotationProxy;
!org.redisson.mapreduce.CoordinatorTask;
!weka.experiment.CSVResultListener;
!org.apache.bval.jsr.util.AnnotationProxy;
!org.jdesktop.swingx.action.ServerAction;
!org.apache.ibatis.logging.jdbc.ResultSetLogger;
!com.alibaba.druid.pool.DruidDataSource;
!org.springframework.core.SerializableTypeWrapper$TypeProxyInvocationHandler;
!org.apache.activemq.jndi.ReadOnlyContext;
!org.apache.camel.util.jndi.JndiContext;
!net.sourceforge.jtds.jdbcx.JtdsDataSource;
!io.lettuce.core.dynamic.support.TypeWrapper$TypeProxyInvocationHandler;
!org.apache.derby.client.BasicClientDataSource;
!com.sun.org.apache.xalan.internal.xsltc.trax.TemplatesImpl;
!org.apache.hadoop.security.UserGroupInformation$AutoRenewalForUserCredsRunnable;
!org.apache.hadoop.security.UserGroupInformation$TicketCacheRenewalRunnable;
!org.eclipse.osgi.internal.url.URLStreamHandlerFactoryImpl;
!org.apache.poi.openxml4j.opc.OPCPackage;
!org.h2.tools.GUIConsole;
!org.apache.hadoop.fs.store.DataBlocks$DiskBlock;
!org.apache.cassandra.io.util.RewindableDataInputStreamPlus;
!org.apache.logging.log4j.core.appender.rolling.action.ZipCompressAction;
!org.apache.openjpa.jdbc.sql.PostgresDictionary;
!org.jgroups.protocols.PDC;
!org.apache.catalina.core.NamingContextListener;
!org.python.core.WrappedIterIterator;
!net.bytebuddy.ClassFileVersion$VersionLocator$Resolver;
!org.apache.commons.beanutils.LazyDynaBean;
!org.springframework.integration.metadata.PropertiesPersistingMetadataStore;
!com.mysql.cj.protocol.ProtocolEntityFactory;
!org.apache.sshd.agent.unix.AgentServerProxy;
!org.apache.calcite.adapter.druid.DruidConnectionImpl;
!org.codehaus.groovy.runtime.ConversionHandler;
!com.google.javascript.jscomp.jar.jar.org.apache.tools.ant.types.resources.URLResource;
!io.undertow.server.handlers.resource.URLResource$1ServerTask;
!com.clickhouse.client.http.HttpURLConnectionImpl;
!liquibase.database.core.DerbyDatabase;
!edu.stanford.nlp.objectbank.ReaderIteratorFactory$ReaderIterator;
!org.eclipse.persistence.internal.databaseaccess.DataSourceAccessor;
!org.hibernate.jpa.spi.NativeQueryConstructorTransformer;

```

Listing 18: Deserialization filter according to verified dormant gadget chains.

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